

Your grade: 100%

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Next item →

1. What property does the general Bayesian recursion derived in this lesson share with the Kalman filter derived in the second course in this specialization?

1 / 1 point

- ☒ The solution for both derivations breaks up into prediction and updating computations for every iteration.
- ☐ Both methods seek to approximate the conditional mean of the state vector, $\mathbb{E}[x_k \mid \mathbb{Z}_k]$.
- ☐ Both methods assume that the state vector has a Gaussian distribution.
- ☐ Both methods assume that the process and sensor noises have Gaussian distributions.

☒ **Correct**
Yes, this is correct.

2. This lesson derived some complicated integral expressions. How will we implement them in the particle filter?

1 / 1 point

- ☒ By using Monte-Carlo integration methods.
- ☐ By assuming that all pdfs are Gaussian and solving the integrals in closed form.
- ☐ By brute force, using a rectangular or triangular integration approximation.
- ☐ By summing all values of the noise pdfs that produce the output whose probability we are seeking to find.

☒ **Correct**
Yes, this is correct.

3. Which metric describing an uncertain state vector x_k contains more information?

1 / 1 point

- ☐ $\mathbb{E}[x_k \mid \mathbb{Z}_k]$.
- ☒ $f(x_k \mid \mathbb{Z}_k)$.

☒ **Correct**
Yes, this is correct. We can compute the conditional mean, variance, mode, median (etc.) from this pdf, so it has more information than simply the conditional mean.